

UNIVERSITY OF CAPE COAST

B.Sc. MICROBIOLOGY THESIS

Fungal Microflora of Indoor Air in Reception Areas of Student Halls of Residence at the University of Cape Coast

A Thesis Submitted to the Department of Microbiology,
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Abstract

Indoor air quality in institutional buildings is a critical determinant of occupant health, particularly in densely populated student accommodation facilities where prolonged exposure to airborne fungal spores may pose significant health risks. This study investigated the fungal microflora of indoor air in the reception areas of six student halls of residence at the University of Cape Coast, Ghana, during the dry season in November. Airborne fungal spores were collected using the settle plate method with Sabouraud Dextrose Agar (SDA) during morning (08:00–10:00) and afternoon (14:00–16:00) sampling periods. Fungal isolates were identified through macroscopic and microscopic morphological characterization, and concentrations were estimated using Omeliansky's formula. Four fungal species were identified: *Aspergillus niger*, *Aspergillus flavus*, *Aspergillus fumigatus*, and *Penicillium chrysogenum*. The estimated fungal concentrations ranged from 21.0 to 49.3 CFU/m³, with all values falling below the World Health Organization (WHO) guideline limit of 50 CFU/m³ for indoor environments. Morning concentrations were consistently higher than afternoon values across all sampling sites, a pattern attributed to the continuous filtration of indoor air by Heating, Ventilation, and Air Conditioning (HVAC) systems operating throughout the day. Antifungal susceptibility testing using the Kirby-Bauer disc diffusion method revealed that all isolates were susceptible to ketoconazole and itraconazole, variably susceptible to griseofulvin, and uniformly resistant to fluconazole. The complete resistance to fluconazole, a first-line azole antifungal, raises significant public health concerns. These findings indicate that while indoor fungal concentrations in the sampled halls are currently within acceptable WHO limits, the presence of pathogenic species and the observed fluconazole resistance warrant ongoing surveillance and proactive indoor air quality management.

Keywords: indoor air quality, fungal aerosols, settle plate method, HVAC systems, antifungal susceptibility, WHO guidelines, *Aspergillus*, *Penicillium*

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1 Introduction

1.1 Background of the Study

Indoor air quality has emerged as one of the most significant environmental health concerns of the twenty-first century, particularly as people in modern societies spend approximately 80–90% of their time indoors[?]. The quality of air within enclosed spaces is influenced by a complex interplay of factors, including ventilation systems, occupancy density, building materials, outdoor air quality, and climatic conditions. Among the various biological contaminants that compromise indoor air quality, fungal spores represent a particularly insidious threat due to their ubiquity in the environment, their capacity for aerial dispersal, and their well-documented associations with a spectrum of adverse health outcomes[?]. Fungi are among the most resilient and adaptable organisms on Earth, capable of colonizing virtually any substrate under suitable environmental conditions. In indoor environments, fungal spores can originate from outdoor air, from fungal growth on building materials, or from human activities that disturb contaminated surfaces.

The reception areas of student halls of residence represent a unique indoor microenvironment worthy of investigation for several reasons. These spaces serve as high-traffic zones where large numbers of students, visitors, and staff congregate daily, creating conditions that may facilitate the accumulation and dispersal of airborne fungal spores. The constant movement of people through entrance doors introduces outdoor air and associated microbial flora into the indoor environment, while the occupancy density itself generates conditions of elevated humidity and temperature that may favour fungal proliferation[?]. Furthermore, in tropical climates such as that of Cape Coast, Ghana, the warm and humid environmental conditions provide an ideal milieu for fungal growth and sporulation, making the monitoring of indoor fungal concentrations particularly relevant in such settings.

The assessment of indoor fungal contamination has traditionally relied on culture-based methods, with the settle plate technique remaining one of the most widely used approaches due to its simplicity, cost-effectiveness, and accessibility in resource-limited settings[?]. The technique involves exposing agar plates to the ambient air for a defined period, allowing airborne fungal spores to settle by gravity onto the culture medium. Although the settle plate method does not provide a direct measure of air volume sampled, estimated fungal concentrations can be derived using established formulae such as Omeliansky's formula, which converts colony counts into estimated concentrations expressed in colony-forming units per cubic metre (CFU/m³)[?]. This metric enables comparison with regulatory standards and guidelines established by organisations such as the World Health Organization (WHO), which has set a recommended limit of 50 CFU/m³ for indoor fungal concentrations[?].

1.2 Statement of the Problem

Indoor air quality in student accommodation facilities at the University of Cape Coast has not been systematically investigated, despite the potential health risks that airborne fungal spores may pose to the large population of students and staff who occupy these buildings daily. Student halls of residence are characterised by high occupancy densities,

continuous foot traffic through reception areas, and environmental conditions that may promote the accumulation and dispersal of fungal bioaerosols. The tropical climate of Cape Coast, with its characteristic warmth and humidity, further exacerbates the potential for indoor fungal contamination, as these conditions are known to support the growth and sporulation of a wide range of fungal species.

Several studies conducted in similar institutional settings across sub-Saharan Africa have reported elevated fungal concentrations in indoor air, with many exceeding the WHO guideline of 50 CFU/m³ [1]. However, the specific conditions prevailing in the reception areas of student halls at the University of Cape Coast—including the operation of Heating, Ventilation, and Air Conditioning (HVAC) systems, the constant opening of main entrance doors, and the building design features—may influence fungal concentrations in ways that differ from other institutional settings. The absence of baseline data on indoor fungal concentrations in these facilities represents a significant gap in the environmental health knowledge base of the university community. Without such data, it is impossible to assess the potential health risks to occupants or to develop evidence-based strategies for improving indoor air quality. This study therefore sought to characterise the fungal microflora of indoor air in the reception areas of student halls of residence at the University of Cape Coast and to determine whether the fungal concentrations in these spaces comply with WHO guidelines.

1.3 Research Objectives

Main Objective: To assess the fungal microflora of indoor air in the reception areas of student halls of residence at the University of Cape Coast.

Specific Objectives:

- (i) To isolate and identify fungal species present in the indoor air of reception areas in selected student halls of residence at the University of Cape Coast.
- (ii) To estimate the concentrations of airborne fungi in the reception areas using the settle plate method and Omeliansky's formula.
- (iii) To compare the estimated fungal concentrations with WHO guidelines for indoor air quality.
- (iv) To determine the temporal variation in fungal concentrations between morning and afternoon sampling periods.
- (v) To evaluate the antifungal susceptibility profiles of the isolated fungal species against selected antifungal agents.

1.4 Research Questions

- (i) What fungal species are present in the indoor air of reception areas in student halls of residence at the University of Cape Coast?
- (ii) What are the estimated concentrations of airborne fungi in these reception areas?

- (iii) Do the estimated fungal concentrations comply with the WHO guideline of 50 CFU/m³ for indoor environments?
- (iv) Is there a significant difference in fungal concentrations between morning and afternoon sampling periods?
- (v) What are the antifungal susceptibility profiles of the isolated fungal species?

1.5 Significance of the Study

This study holds considerable significance for both the academic community and the broader field of environmental health. First, it provides baseline data on the fungal composition and concentration of indoor air in student halls of residence at the University of Cape Coast, filling a critical gap in the existing knowledge base. Such baseline data are essential for establishing reference values against which future monitoring efforts can be compared, enabling the detection of changes in indoor air quality over time. Second, the findings of this study contribute to the growing body of literature on indoor air quality in tropical institutional settings, a context that remains underrepresented in the global research landscape. Third, by comparing estimated fungal concentrations with WHO guidelines, this study provides an evidence-based assessment of the health risks associated with indoor fungal exposure in the sampled facilities, thereby informing university administrators and health authorities about the need for, or adequacy of, current indoor air quality management practices.

Furthermore, the antifungal susceptibility data generated by this study have direct clinical relevance, as they provide information about the drug sensitivity patterns of environmental fungal isolates, which may inform treatment decisions in cases of fungal infection among the student population. The identification of antifungal resistance among environmental isolates is of particular importance in the context of the global antimicrobial resistance crisis, as it highlights potential challenges in the clinical management of fungal infections acquired from the indoor environment. Finally, the methodological approach employed in this study—combining settle plate sampling, morphological identification, concentration estimation using Omeliansky’s formula, and antifungal susceptibility testing—serves as a replicable model for similar investigations in other institutional settings across Ghana and the wider West African sub-region.

1.6 Scope and Limitations

This study was delimited to the reception areas of six student halls of residence at the University of Cape Coast, namely Volta Hall, Limpopo Hall, Atlantic Hall, Kwame Nkrumah Hall, Adehye Hall, and Valco Hall. The study focused exclusively on the assessment of culturable airborne fungi using the settle plate method with Sabouraud Dextrose Agar (SDA) as the culture medium. Sampling was conducted during the dry season (November), and the results may not be representative of fungal concentrations during other seasons, particularly the wet season when humidity levels are substantially higher. Fungal identification was based on macroscopic and microscopic morphological characteristics, which, while well-established and widely used, may not provide the same level of taxonomic resolution as molecular methods such as DNA sequencing. The settle plate

method, although practical and cost-effective, has inherent limitations as a passive sampling technique: it does not measure a known volume of air, relies on gravitational settling of spores, and may underestimate the concentrations of small spores that remain airborne for extended periods. These limitations are partially addressed by the application of Omeliansky's formula, which provides an estimated concentration based on established conversion factors. Additionally, the study assessed antifungal susceptibility against only four antifungal agents, and the results may not reflect susceptibility patterns to other available antifungal drugs.

2 Literature Review

2.1 Indoor Air Quality: An Overview

Indoor air quality (IAQ) refers to the chemical, physical, and biological characteristics of air within and around buildings and structures, particularly as they relate to the health and comfort of building occupants[?]. The significance of IAQ has been increasingly recognised in recent decades, driven by the growing awareness that indoor environments can harbour a diverse array of pollutants that may exceed outdoor levels by factors of two to five, and in some cases by orders of magnitude[?]. The World Health Organization estimates that approximately 3.8 million premature deaths annually are attributable to household air pollution, underscoring the critical importance of monitoring and improving the quality of air in enclosed spaces.

The biological component of indoor air pollution encompasses a wide variety of microorganisms, including bacteria, viruses, fungal spores, pollen, and fragments of biological origin such as endotoxins and mycotoxins[?]. Among these, fungal spores are particularly noteworthy due to their small size (typically 2–20 μm), their ability to remain airborne for extended periods, and their capacity to trigger allergic reactions, respiratory symptoms, and invasive infections in susceptible individuals. The assessment of indoor fungal contamination is complicated by the dynamic nature of fungal populations, which fluctuate in response to changes in temperature, humidity, ventilation, occupancy, and seasonal factors. Consequently, accurate evaluation of IAQ with respect to fungal contamination requires systematic sampling under defined conditions, combined with appropriate analytical methods and reference standards.

The concept of “sick building syndrome” (SBS) has been closely linked to poor indoor air quality, with fungal contamination identified as one of the primary contributing factors[?]. Symptoms associated with SBS include mucosal irritation, headaches, fatigue, and respiratory complaints that are temporally related to occupancy of a particular building and that improve upon leaving the building. While the aetiology of SBS is multifactorial, the presence of elevated fungal concentrations in indoor air has been consistently associated with increased symptom prevalence, highlighting the need for routine monitoring of fungal bioaerosols in institutional and residential buildings.

2.2 Fungal Aerosols and Their Sources

Fungal aerosols, also referred to as fungal bioaerosols, consist of airborne fungal propagules, including spores, conidia, hyphal fragments, and spore-bearing structures, that are capable of remaining suspended in the atmosphere for variable periods depending on their size, morphology, and environmental conditions[?]. The primary sources of fungal aerosols in indoor environments include outdoor air infiltrating through doors, windows, and ventilation systems; indoor fungal growth on damp or water-damaged building materials such as ceiling tiles, wallpaper, and carpet; and human activities that disturb contaminated surfaces, such as cleaning, renovation, or the movement of occupants[?].

The genus *Aspergillus* is among the most frequently isolated fungal genera in indoor air studies worldwide, reflecting its cosmopolitan distribution and its ability to grow on a wide

range of substrates under diverse environmental conditions[?]. *Aspergillus niger*, *A. flavus*, and *A. fumigatus* are commonly reported indoor air isolates, each with distinct ecological preferences and clinical significance. *Aspergillus niger* is a ubiquitous saprophyte found in soil and decomposing vegetation, and while it is considered an opportunistic pathogen, it is the most frequently isolated *Aspergillus* species in indoor air surveys. *Aspergillus flavus* is of particular concern due to its production of aflatoxins, potent carcinogenic mycotoxins that pose a serious threat to human health through both inhalation and ingestion routes. *Aspergillus fumigatus* is the most pathogenic member of the genus and the leading cause of invasive aspergillosis, a life-threatening infection primarily affecting immunocompromised individuals.

The genus *Penicillium* is another dominant component of indoor fungal aerosols, with *P. chrysogenum* being one of the most commonly encountered species[?]. *Penicillium* species are prolific spore producers, and their small conidia (typically 2.5–5 μm) are efficiently dispersed in indoor air, where they can remain suspended for extended periods. While *Penicillium* species are generally considered less pathogenic than *Aspergillus* species, they are significant allergens and have been implicated in cases of hypersensitivity pneumonitis and occupational asthma. The production of mycotoxins by certain *Penicillium* species further adds to their potential health significance in indoor environments.

2.3 Health Implications of Indoor Fungal Exposure

Exposure to airborne fungal spores in indoor environments is associated with a wide spectrum of health effects, ranging from mild allergic reactions to life-threatening invasive infections[?]. The health consequences of fungal exposure depend on multiple factors, including the species and concentration of fungal spores, the duration and frequency of exposure, the immune status of the exposed individual, and the presence of co-existing environmental stressors. In immunocompetent individuals, the most common health effects of indoor fungal exposure are allergic reactions, including allergic rhinitis, conjunctivitis, and asthma exacerbation. Fungal allergens act as sensitising agents that trigger immunoglobulin E (IgE)-mediated hypersensitivity responses in genetically predisposed individuals, leading to the characteristic symptoms of sneezing, nasal congestion, watery eyes, and bronchial constriction[?].

In immunocompromised individuals, including patients undergoing chemotherapy, organ transplant recipients, and individuals with advanced HIV infection, exposure to opportunistic fungal pathogens such as *Aspergillus fumigatus* can result in invasive fungal infections with high mortality rates. Invasive aspergillosis, for example, carries a mortality rate of 40–90% depending on the patient population and the timeliness of diagnosis and treatment[?]. The inhalation of *A. fumigatus* conidia allows the fungus to colonise the respiratory tract, from where it can invade pulmonary blood vessels and disseminate to other organs, including the brain, heart, and kidneys.

Beyond allergic and infectious outcomes, exposure to indoor fungi has been associated with toxic effects resulting from the inhalation of mycotoxins, secondary metabolites produced by various fungal species during their growth and sporulation. Aflatoxins produced by *A. flavus* and ochratoxin A produced by certain *Aspergillus* and *Penicillium* species are among the most potent mycotoxins, with well-established carcinogenic, nephrotoxic, and hepatotoxic properties[?]. While the majority of mycotoxin exposure occurs through the

ingestion of contaminated food, the inhalation of mycotoxin-containing spores in indoor environments represents an additional route of exposure that has been less thoroughly investigated but may pose significant health risks, particularly in environments with high fungal concentrations.

2.4 Climatic and Seasonal Influences on Indoor Fungal Concentrations

Climatic and seasonal factors exert a profound influence on the concentration and composition of fungal aerosols in both outdoor and indoor environments⁷. Temperature and humidity are the two primary climatic variables that determine fungal growth rates, sporulation intensity, and spore release dynamics. In tropical and subtropical regions, the warm and humid conditions that prevail for much of the year create an environment that is highly conducive to fungal growth and reproduction, resulting in consistently elevated outdoor fungal spore counts that can serve as a continuous source of indoor contamination through the infiltration of outdoor air.

The study was conducted during the dry season (November), prior to the onset of the harmattan dust-laden trade winds. Sampling occurred during a period characterised by relatively low humidity (65–75%) and moderate temperatures (28–32°C), conditions typical of the pre-harmattan transitional period in the coastal savanna zone of Cape Coast, Ghana. During this transitional period, the prevailing climatic conditions support moderate fungal growth and sporulation, while the absence of the extreme desiccation associated with the harmattan winds means that fungal populations remain viable and capable of aerial dispersal. Studies conducted in similar West African coastal environments have demonstrated that fungal concentrations in indoor air during the dry season typically fall between the elevated levels observed during the wet season and the reduced levels associated with the peak harmattan period⁷.

The influence of climatic factors on indoor fungal concentrations is mediated by several mechanisms. Elevated outdoor temperatures increase the rate of fungal metabolism and sporulation, leading to higher outdoor spore counts that can infiltrate indoor environments through ventilation systems and open doors. Humidity affects both the growth of fungi on indoor surfaces and the release of spores from fungal colonies; many fungal species exhibit enhanced spore release under conditions of rapid changes in humidity, a phenomenon known as “wet-dry discharge.” The interaction between temperature and humidity is particularly important, as the combination of warm temperatures and high relative humidity creates the most favourable conditions for fungal proliferation in indoor environments.

2.5 The Role of HVAC Systems in Indoor Air Quality

Heating, Ventilation, and Air Conditioning (HVAC) systems are mechanical systems designed to control indoor environmental conditions by regulating temperature, humidity, and air quality. In the context of indoor air quality, HVAC systems play a critical role in filtering airborne particulates, including fungal spores, through their mechanical filtration components. The filters within HVAC units—typically consisting of pre-filters and high-efficiency particulate air (HEPA) filters in more advanced systems—physically trap spores

as air is drawn through the system for conditioning and recirculation, thereby reducing the concentration of viable fungal propagules in the indoor environment[?].

The effectiveness of HVAC systems in controlling indoor fungal concentrations depends on several factors, including the type and efficiency of the filters installed, the rate of air exchange, the maintenance schedule for filter replacement, and the overall design of the ventilation system. Properly maintained HVAC systems with appropriately rated filters can significantly reduce the concentration of airborne fungal spores, with studies reporting reductions of 50–90% in fungal concentrations in buildings with well-functioning HVAC systems compared to naturally ventilated buildings[?]. The continuous operation of HVAC systems throughout the day can lead to a progressive reduction in indoor fungal concentrations, as the recirculated air passes through the filtration media multiple times, with each pass removing a fraction of the remaining airborne spores.

However, HVAC systems can also serve as sources of fungal contamination if not properly maintained. Condensation within ductwork, on cooling coils, and in drip pans can create moisture-rich environments that support fungal growth, potentially leading to the dispersal of fungal spores throughout the building via the ventilation system. This phenomenon, sometimes referred to as “HVAC-associated fungal contamination,” has been documented in numerous studies and represents a significant concern for building managers and public health officials. Regular inspection, cleaning, and maintenance of HVAC components are therefore essential to ensure that these systems fulfil their intended function of improving indoor air quality rather than contributing to its deterioration.

2.6 Antifungal Susceptibility and Resistance

Antifungal susceptibility testing is a laboratory procedure used to determine the sensitivity of fungal isolates to specific antifungal agents, providing critical information for the clinical management of fungal infections[?]. The Kirby-Bauer disc diffusion method is one of the most widely used techniques for antifungal susceptibility testing, offering a simple, reproducible, and cost-effective approach that is particularly suitable for resource-limited settings. In this method, filter paper discs impregnated with known concentrations of antifungal agents are placed on agar plates inoculated with the test organism, and the zone of inhibition—the clear area around each disc where fungal growth has been suppressed—is measured after a defined incubation period[?].

The emergence of antifungal resistance among environmental fungal isolates is a growing concern in the context of global antimicrobial resistance. Resistance to the azole class of antifungal agents, which includes fluconazole, itraconazole, ketoconazole, and voriconazole, has been increasingly reported in both clinical and environmental isolates of *Aspergillus* species[?]. The mechanisms of azole resistance include mutations in the *cyp51A* gene encoding the target enzyme lanosterol 14 α -demethylase, overexpression of efflux pumps, and the induction of alternative metabolic pathways. Of particular concern is the observation that azole-resistant *Aspergillus* isolates have been recovered from environmental samples in the absence of prior clinical azole exposure, suggesting that the widespread agricultural use of azole fungicides may be driving the selection of resistant strains in the environment[?].

The clinical implications of antifungal resistance are profound. Fluconazole, one of the most commonly prescribed antifungal agents, has been a mainstay of treatment for

candidiasis and other fungal infections for decades due to its favourable pharmacokinetic profile, excellent oral bioavailability, and low toxicity. However, the emergence of fluconazole resistance among environmental *Aspergillus* and *Penicillium* isolates limits the therapeutic options available for treating infections caused by these organisms, necessitating the use of alternative agents such as amphotericin B (which carries a significant risk of nephrotoxicity) or the newer echinocandin class (which must be administered intravenously). Monitoring antifungal susceptibility patterns among environmental fungal isolates is therefore essential for informing treatment guidelines and for detecting the emergence of resistance at an early stage.

2.7 Regulatory Standards for Indoor Fungal Concentrations

Regulatory standards for indoor fungal concentrations vary across jurisdictions, but the most widely referenced guideline is the World Health Organization's recommendation that indoor fungal concentrations should not exceed 50 CFU/m³[?]. This guideline is based on the collective evidence from epidemiological studies demonstrating that indoor environments with fungal concentrations exceeding this threshold are associated with an increased risk of adverse health effects among occupants. The WHO guideline is intended as a general reference value rather than a strict regulatory limit, as the health significance of a given fungal concentration depends on the species composition, the susceptibility of the exposed population, and the duration of exposure.

Several countries and organisations have established their own indoor air quality standards for fungal contamination. In the United States, the Occupational Safety and Health Administration (OSHA) has not established a specific permissible exposure limit for fungal bioaerosols, but the American Conference of Governmental Industrial Hygienists (ACGIH) has recommended that indoor fungal concentrations should be significantly lower than outdoor concentrations and that the presence of indicator species such as *Stachybotrys chartarum* or *Aspergillus fumigatus* should trigger investigation regardless of the measured concentration[?]. In Europe, the European Commission's INDEX project recommended a limit of 50 CFU/m³ for indoor fungal concentrations, consistent with the WHO guideline, and further recommended that the identification of fungal species should accompany concentration measurements to provide a more comprehensive assessment of health risks.

The interpretation of indoor fungal concentration data in the context of regulatory standards requires careful consideration of several factors. First, the sampling method employed can significantly influence the measured concentrations; active sampling methods that draw a known volume of air through a collection device typically yield higher and more reproducible values than passive methods such as the settle plate technique. Second, the temporal variability of fungal concentrations means that single-point measurements may not be representative of the long-term exposure conditions experienced by building occupants. Third, the species composition of the fungal aerosol is at least as important as the total concentration, as pathogenic species such as *Aspergillus fumigatus* pose a greater health risk at a given concentration than do non-pathogenic species.

3 Materials and Methods

3.1 Study Area and Climatic Conditions

The study was conducted at the University of Cape Coast, located in the Central Region of Ghana, approximately 165 km west of Accra. The university campus is situated in the coastal savanna zone, characterised by a tropical climate with distinct wet and dry seasons. The study was conducted during the dry season (November), prior to the onset of the harmattan dust-laden trade winds. Sampling occurred during a period characterised by relatively low humidity (65–75%) and moderate temperatures (28–32°C), conditions typical of the pre-harmattan transitional period in the coastal savanna zone of Cape Coast, Ghana. The coastal location of the university means that ambient humidity levels remain moderately high even during the dry season, creating conditions that support moderate fungal growth and sporulation activity.

Six student halls of residence were selected for this study: Volta Hall, Limpopo Hall, Atlantic Hall, Kwame Nkrumah Hall, Adehye Hall, and Valco Hall. These halls were selected to represent a range of building designs, ages, and occupancy levels within the university's accommodation system. The reception areas of these halls share certain common features: they serve as the primary entry and exit points for residents and visitors, they experience high foot traffic throughout the day, and they are equipped with Heating, Ventilation, and Air Conditioning (HVAC) systems that operate continuously to maintain comfortable indoor temperatures and humidity levels. The entrance doors of all sampled halls are wide open at all times during the day, and windows are permanently closed because the air conditioning units are constantly on.

3.2 Study Design and Period

This study employed a cross-sectional descriptive design to assess the fungal microflora of indoor air in the reception areas of the selected student halls of residence. The cross-sectional design was appropriate for this investigation as it enabled the collection of data at a specific point in time, providing a snapshot of the fungal composition and concentration in the sampled environments. Sampling was conducted over a two-week period in November, with each hall sampled on two occasions (morning and afternoon) during the study period. The morning sampling period was between 08:00 and 10:00 GMT, while the afternoon sampling period was between 14:00 and 16:00 GMT. These time windows were selected to capture potential diurnal variation in fungal concentrations associated with changes in occupancy patterns, HVAC operation duration, and the progressive filtration of indoor air throughout the day.

The sampling protocol involved the placement of triplicate settle plates at each sampling site during each sampling period, yielding a total of 36 plate exposures across the six halls (6 halls $\times$ 2 periods $\times$ 3 replicates). The settle plates were positioned at approximately 1 metre above floor level, which corresponds to the breathing zone of seated and standing occupants, and were placed away from direct drafts from the HVAC vents and the main entrance to avoid localised effects on spore deposition. All plates were exposed for a standard period of 15 minutes, after which they were immediately sealed, labelled, and transported to the laboratory for incubation and analysis.

3.3 Sample Collection (Settle Plate Method)

Airborne fungal spores were collected using the settle plate method, a passive sampling technique that relies on the gravitational settling of particles onto an exposed agar surface. Sabouraud Dextrose Agar (SDA) supplemented with chloramphenicol (50 mg/L) was used as the culture medium to inhibit bacterial growth while permitting the growth of fungi. The SDA plates were prepared according to the manufacturer's instructions, autoclaved at 121°C for 15 minutes, and poured into sterile 90 mm Petri dishes under aseptic conditions in a laminar flow cabinet. After solidification, the plates were pre-incubated at 25°C for 24 hours to check for contamination before use in the field.

During sampling, the lids of the Petri dishes were removed and the agar surfaces were exposed to the ambient air for 15 minutes at each sampling location. Three replicate plates were used at each site during each sampling period to improve the precision of the colony count estimates. After exposure, the plates were immediately covered, sealed with parafilm, labelled with the sampling site, date, and time, and transported to the Microbiology Laboratory of the University of Cape Coast in insulated containers. Control plates (unexposed SDA plates opened briefly at the laboratory) were included in each batch of sampling to account for any contamination that might occur during plate preparation, transport, or handling.

3.4 Culture and Identification of Fungal Isolates

Following sample collection, the exposed settle plates were incubated at 25°C for 5–7 days in an inverted position to allow fungal growth and colony development. After incubation, colonies were counted and recorded for each plate. The colony counts from the triplicate plates at each sampling site were averaged to obtain the mean colony count for each sampling period. Fungal colonies were initially differentiated based on their macroscopic characteristics, including colony colour, texture, surface morphology, and reverse pigmentation. Distinct colony types were subcultured onto fresh SDA plates to obtain pure cultures for detailed identification.

Identification of fungal isolates was performed using a combination of macroscopic and microscopic morphological examination. Macroscopic characters assessed included colony colour, texture (velvety, powdery, cottony, or granular), diameter, presence of exudate, and reverse pigmentation on SDA. Microscopic examination was conducted using the lactophenol cotton blue (LPCB) wet mount technique, in which a small portion of the fungal colony was teased apart using sterile needles, mounted in LPCB staining solution on a glass slide, and examined under the light microscope at $\times 100$, $\times 400$, and $\times 1000$ magnification. Key microscopic features evaluated included the morphology and arrangement of conidiophores, the shape and size of conidia, the presence and structure of vesicles and phialides (for *Aspergillus* species), and the arrangement of metulae and phialides (for *Penicillium* species). Identification was performed using standard mycological keys and reference texts^{??}.

3.5 Estimation of Fungal Concentrations (Omeliensky's Formula)

The estimation of airborne fungal concentrations from settle plate data was performed using Omeliensky's formula, which converts settle plate colony counts into estimated concentrations expressed in colony-forming units per cubic metre (CFU/m³). The formula is expressed as follows:

$$\text{CFU/m}^3 = \frac{n \times 10,000}{\pi \times r^2 \times t} \quad (1)$$

where n is the number of colonies counted on the plate, r is the radius of the Petri dish in centimetres (4.5 cm for a standard 90 mm plate), and t is the exposure time in minutes (15 minutes in this study). The factor 10,000 converts the result from colonies per square centimetre-minute to the standard unit of CFU/m³, accounting for the relationship between gravitational settling rate and the assumed particle deposition velocity. The conversion factor derived from the formula under the conditions used in this study is approximately 10.48, such that the estimated concentration in CFU/m³ equals the average colony count multiplied by 10.48.

This formula provides an approximate estimate of fungal concentrations and is based on the assumption that the settling velocity of fungal spores follows Stokes' law for spherical particles in still air. While the settle plate method and Omeliensky's formula have well-recognised limitations—including the inability to account for variations in spore size, density, and aerodynamic behaviour, as well as the effects of air currents on spore deposition—they remain widely used in resource-limited settings and provide a standardised approach for comparing results across studies and against regulatory guidelines.

3.6 Antifungal Susceptibility Testing (Kirby-Bauer Disc Diffusion)

Antifungal susceptibility testing was performed using the Kirby-Bauer disc diffusion method in accordance with the guidelines established by the Clinical and Laboratory Standards Institute (CLSI) document M44-A2⁷. Pure cultures of each identified fungal isolate were used to prepare spore suspensions in sterile saline (0.85% NaCl) adjusted to a turbidity equivalent to 0.5 McFarland standard, corresponding to approximately $1\text{--}5 \times 10^6$ colony-forming units per millilitre. The spore suspensions were swabbed onto the surface of Mueller-Hinton Agar supplemented with 2% glucose and 0.5 µg/mL methylene blue (MHGM) to enhance the growth and detection of antifungal activity.

Four antifungal agents were tested: ketoconazole (10 µg), fluconazole (25 µg), itraconazole (10 µg), and griseofulvin (25 µg). Antifungal discs were placed on the inoculated agar surfaces using sterile forceps, with a minimum inter-disc distance of 24 mm to prevent overlap of inhibition zones. The plates were incubated at 25°C for 48–72 hours, after which the diameters of the zones of inhibition were measured in millimetres using a calibrated ruler. The results were interpreted according to CLSI breakpoint criteria as susceptible (S), intermediate (I), or resistant (R) based on the measured zone diameters for each antifungal agent–organism combination. Quality control was maintained

by testing reference strains of *Candida albicans* (ATCC 90028) concurrently with the test organisms to ensure the reliability and reproducibility of the results.

4 Results

4.1 Total Fungal Colony Counts from Settle Plates

The fungal colony counts obtained from the settle plate exposures in the reception areas of the six student halls of residence are presented in Table 1. Triplicate plates were exposed at each sampling site during both the morning (08:00–10:00) and afternoon (14:00–16:00) sampling periods, and the mean colony count was calculated for each site and period. Morning colony counts ranged from 2.0 to 4.7 across all sites, while afternoon counts ranged from 2.0 to 2.7. In all cases, morning counts were higher than the corresponding afternoon values, with differences ranging from 0.7 to 2.4 colonies per plate on average.

Table 1: Colony counts from settle plate exposures in reception areas of student halls of residence (triplicate plates per sampling period).

Sampling Site	AM Colony Counts			AM Avg	PM Colony Counts			PM Avg
	P1	P2	P3		P1	P2	P3	
Volta Hall	5	4	4	4.3	3	3	2	2.7
Limpopo Hall	5	4	5	4.7	3	2	3	2.7
Atlantic Hall	4	4	3	3.7	2	3	2	2.3
Kwame Nkrumah Hall	4	5	4	4.3	3	2	2	2.3
Adehye Hall	4	3	4	3.7	2	2	3	2.3
Valco Hall	3	4	4	3.7	2	2	2	2.0

The highest mean morning colony count was recorded at Limpopo Hall (4.7), followed by Volta Hall and Kwame Nkrumah Hall (both 4.3). The lowest mean morning count was recorded at Atlantic Hall, Adehye Hall, and Valco Hall (3.7 each). In the afternoon, the highest mean colony count was observed at Volta Hall and Limpopo Hall (2.7 each), while the lowest was at Valco Hall (2.0). The consistent reduction in colony counts from morning to afternoon across all sites suggests that a common mechanism is responsible for the diurnal variation observed in fungal concentrations.

4.2 Identification of Fungal Isolates

Four fungal species were identified from the air samples collected in the reception areas of the student halls of residence. These comprised three species of the genus *Aspergillus*—*Aspergillus niger*, *Aspergillus flavus*, and *Aspergillus fumigatus*—and one species of the genus *Penicillium*—*Penicillium chrysogenum*. All four species were isolated from every sampling site, although their relative frequencies varied between sites and between the morning and afternoon sampling periods. The predominance of *Aspergillus* species among the isolated fungi is consistent with the findings of previous indoor air quality studies conducted in tropical and subtropical regions, where members of this genus are consistently reported as the most frequently isolated fungal contaminants of indoor air.

The identification of *Aspergillus* species was based on the characteristic morphology of their conidiophores, which terminate in swollen vesicles bearing phialides that produce chains of conidia. The differentiation of the three *Aspergillus* species was achieved through the examination of colony morphology on SDA and the detailed assessment of microscopic

features, including the size and shape of vesicles, the arrangement of phialides (uniseriate vs. biseriate), the colour and ornamentation of conidia, and the presence or absence of specialised structures such as Hulle cells. *Penicillium chrysogenum* was identified by its characteristic brush-like conidiophores (penicilli) bearing chains of small, globose conidia, and by the blue-green colony colouration with a white margin on SDA.

4.3 Morphological Characterization

4.3.1 Microscopic Morphology

The microscopic morphology of the four identified fungal species, as observed in lactophenol cotton blue wet mount preparations, is summarised below.

Aspergillus niger: Microscopic examination revealed hyaline, septate hyphae with uniseriate conidiophores terminating in globose vesicles (50–75 μm in diameter). Phialides covered the entire surface of the vesicle and bore chains of dark brown to black, globose conidia (3–5 μm in diameter) with a rough, echinulate surface ornamentation. The conidial chains were densely packed, forming the characteristic radiate head structure.

Aspergillus flavus: Microscopic examination showed hyaline, septate hyphae with both uniseriate and biseriate conidiophores. Vesicles were globose to subglobose (25–45 μm in diameter), with phialides covering the upper two-thirds to three-quarters of the vesicle surface. Conidia were globose to subglobose, pale green, smooth to finely roughened, and measured 3–6 μm in diameter.

Aspergillus fumigatus: Microscopic examination demonstrated hyaline, septate hyphae with uniseriate conidiophores bearing columnar conidial heads. Vesicles were flask-shaped to subglobose (20–30 μm in diameter), with phialides on the upper two-thirds of the vesicle. Conidia were globose to subglobose, green, smooth, and measured 2–3.5 μm in diameter. The columnar arrangement of the conidial chains is a distinguishing feature of this species.

Penicillium chrysogenum: Microscopic examination revealed hyaline, septate hyphae with brush-like conidiophores (penicilli) consisting of smooth-walled stipes bearing terminal biverticillate or irregularly branched structures with metulae and phialides. Conidia were globose to subglobose, smooth-walled, pale green, and measured 2.5–4 μm in diameter. The conidia were borne in disordered chains from the phialides, forming the characteristic brush-like appearance.

4.3.2 Macroscopic Morphology of Pure Cultures

The macroscopic morphology of the pure cultures of the four identified fungal species grown on Sabouraud Dextrose Agar (SDA) is described below and illustrated with image placeholders for Plates 1–4.

Plate 1: *Aspergillus niger* — Colonies on SDA appear initially white, becoming densely black with a white margin; reverse is colorless to pale yellow; texture is powdery to granular. The colony surface is typically flat or slightly raised, with a dense central area of black sporulation that obscures the underlying mycelium. The white margin represents the area of actively growing hyphae that have not yet begun sporulation. Growth is

[IMAGE PLACEHOLDER: Plate 1 — *Aspergillus niger* pure culture on SDA. Colonies on SDA appear initially white, becoming densely black with a white margin; reverse is colorless to pale yellow; texture is powdery to granular.]

moderately rapid, with colonies reaching 4–6 cm in diameter after 7 days of incubation at 25°C.

[IMAGE PLACEHOLDER: Plate 2 — *Aspergillus flavus* pure culture on SDA. Colonies on SDA are yellow-green with a fluffy texture; reverse is gold to reddish-brown; rapidly growing and spreading.]

Plate 2: *Aspergillus flavus* — Colonies on SDA are yellow-green with a fluffy texture; reverse is gold to reddish-brown; rapidly growing and spreading. The colonies are typically granular or powdery in texture, with a more floccose appearance compared to *A. niger*. Growth is rapid, with colonies often reaching 6–7 cm in diameter after 7 days. Some isolates produce sclerotia, which appear as hard, dark bodies embedded in the agar surface. The yellow-green colouration intensifies as sporulation progresses, and the reverse side of the colony develops a characteristic gold to reddish-brown pigmentation.

[IMAGE PLACEHOLDER: Plate 3 — *Aspergillus fumigatus* pure culture on SDA. Colonies on SDA are bluish-green with a smooth to velvety texture; reverse is white to tan; thermotolerant species.]

Plate 3: *Aspergillus fumigatus* — Colonies on SDA are bluish-green with a smooth to velvety texture; reverse is white to tan; thermotolerant species. The colony surface is typically suede-like in texture, with a distinctive bluish-green colouration that is paler than the dark green of *A. flavus* or the black of *A. niger*. Growth is moderately rapid at 25°C but is markedly enhanced at 37°C, reflecting the thermotolerant nature of this species. Colonies reach 3–5 cm in diameter after 7 days at 25°C. The absence of floccose aerial hyphae and the velvety texture help distinguish *A. fumigatus* from other *Aspergillus* species.

[IMAGE PLACEHOLDER: Plate 4 — *Penicillium chrysogenum* pure culture on SDA. Colonies on SDA are blue-green with a white margin; texture is velvety to powdery; reverse is pale yellow to pinkish; produces characteristic brush-like conidiophores.]

Plate 4: *Penicillium chrysogenum* — Colonies on SDA are blue-green with a white margin; texture is velvety to powdery; reverse is pale yellow to pinkish; produces characteristic brush-like conidiophores. The colony surface is flat to slightly raised, with a velvety to powdery texture and a distinctive blue-green colouration that is surrounded by a narrow white margin of non-sporulating hyphae. Growth is moderately rapid, with colonies reaching 3–5 cm in diameter after 7 days at 25°C. The reverse side typically develops a pale yellow to pinkish pigmentation. The production of characteristic brush-like conidiophores is readily observed under low-power microscopy and is a defining feature of the genus *Penicillium*.

4.4 Species Composition and Prevalence

The species composition and prevalence of fungal isolates across the six sampling sites are presented in Table 2. *Aspergillus niger* was the most prevalent species, isolated from all six sampling sites and accounting for the largest proportion of the total isolates. *Aspergillus flavus* and *Aspergillus fumigatus* were also isolated from all six sites, though at slightly lower frequencies. *Penicillium chrysogenum* was isolated from five of the six sampling sites, being absent from the Valco Hall reception area in the afternoon samples.

The overall species composition was dominated by members of the genus *Aspergillus*, which collectively accounted for the majority of fungal isolates recovered from all sampling sites. This finding is consistent with the well-established predominance of *Aspergillus* species in indoor air in tropical environments, where the warm and humid conditions favour the growth and sporulation of these opportunistic fungi. The presence of all three *Aspergillus* species at every sampling site suggests that these fungi are common components of the indoor mycobiota in the student halls of residence, likely originating from

Table 2: Species composition and prevalence of fungal isolates across sampling sites.

Species	Volta	Limpopo	Atlantic	K. Nkrumah	Adehye	Valco	Total
<i>A. niger</i>	+	+	+	+	+	+	6/6
<i>A. flavus</i>	+	+	+	+	+	+	6/6
<i>A. fumigatus</i>	+	+	+	+	+	+	6/6
<i>P. chrysogenum</i>	+	+	+	+	+	—	5/6

outdoor sources and becoming established in the indoor environment through continuous infiltration of outdoor air.

4.5 Estimated Fungal Concentrations (CFU/m³)

The estimated fungal concentrations, calculated from the mean colony counts using Omeliansky's formula, are presented in Table 3. All estimated concentrations fell below the WHO guideline limit of 50 CFU/m³ for indoor environments, indicating that the fungal burden in the reception areas of all six student halls was within acceptable limits at the time of sampling. The overall mean fungal concentrations across the six sampling sites ranged from 29.9 CFU/m³ (Valco Hall) to 38.8 CFU/m³ (Limpopo Hall).

Table 3: Estimated fungal concentrations (CFU/m³) in reception areas of student halls of residence, calculated using Omeliansky's formula.

Sampling Site	AM CFU/m ³	PM CFU/m ³	Overall Mean
Volta Hall	45.1	28.3	36.7
Limpopo Hall	49.3	28.3	38.8
Atlantic Hall	38.8	24.1	31.4
Kwame Nkrumah Hall	45.1	24.1	34.6
Adehye Hall	38.8	24.1	31.4
Valco Hall	38.8	21.0	29.9

The highest estimated fungal concentration was recorded at Limpopo Hall during the morning sampling period (49.3 CFU/m³), which represents the maximum value observed in this study. Notably, even this highest recorded value remains below the WHO guideline of 50 CFU/m³, confirming that all sampled buildings are in compliance with the international standard for indoor air quality. The lowest estimated concentration was observed at Valco Hall during the afternoon sampling period (21.0 CFU/m³). The range of overall mean concentrations across the six sites (29.9–38.8 CFU/m³) indicates relatively uniform fungal burden across the sampled buildings, with no individual hall exhibiting concentrations that would be considered elevated relative to the WHO benchmark.

4.6 Temporal Variation in Fungal Concentrations

A consistent pattern of temporal variation in fungal concentrations was observed across all six sampling sites, with morning concentrations being uniformly higher than afternoon values. The magnitude of the morning-to-afternoon reduction ranged from 16.8 CFU/m³

(Volta Hall) to 21.0 CFU/m³ (Valco Hall and Kwame Nkrumah Hall), with an average reduction of approximately 18.7 CFU/m³ across all sites. This consistent diurnal pattern suggests that a common mechanism is responsible for the observed reduction in fungal concentrations during the afternoon hours.

The reduction in fungal concentrations from morning to afternoon can be attributed to the continuous filtration of indoor air by the Heating, Ventilation, and Air Conditioning (HVAC) systems operating in the reception areas of all six student halls. The entrance doors of these halls are wide open at all times, and windows are permanently closed because the air conditioning units are constantly on. The HVAC systems' mechanical filters progressively trap airborne fungal spores throughout the day, and the continuous air recirculation through filter media reduces spore concentrations in the indoor environment. Over the course of the day, as the HVAC system operates continuously, each pass of the indoor air through the filtration system removes a fraction of the remaining airborne spores, resulting in a cumulative reduction in the indoor fungal load by the afternoon sampling period.

Additionally, the dilution effect contributed by the wide-open main entrance doors allows for the exchange of air between the indoor and outdoor environments. While this exchange can introduce outdoor fungal spores into the indoor environment, the net effect during the afternoon is a reduction in indoor fungal concentrations, likely because the outdoor spore levels during the afternoon are relatively low compared to the morning, and the HVAC filtration effect predominates over the influx of outdoor spores.

4.7 Spatial Variation Across Sampling Sites

The spatial variation in fungal concentrations across the six sampling sites is summarised in Table 3. While all sites exhibited fungal concentrations within the WHO guideline of 50 CFU/m³, some inter-site differences were observed. Limpopo Hall recorded the highest overall mean concentration (38.8 CFU/m³), followed by Volta Hall (36.7 CFU/m³) and Kwame Nkrumah Hall (34.6 CFU/m³). Atlantic Hall and Adehye Hall shared the same overall mean concentration (31.4 CFU/m³), while Valco Hall recorded the lowest overall mean concentration (29.9 CFU/m³).

The observed spatial variation in fungal concentrations may be attributed to differences in occupancy levels, the volume of the reception areas, the efficiency and maintenance status of the HVAC systems, and the orientation of the buildings relative to potential outdoor sources of fungal spores such as vegetation, construction sites, or waste disposal areas. Limpopo Hall, which recorded the highest morning concentration (49.3 CFU/m³), may experience higher morning occupancy or may be located in closer proximity to outdoor fungal sources. Conversely, Valco Hall, which consistently recorded the lowest concentrations, may benefit from a more effective HVAC filtration system or may be situated in a location with fewer outdoor fungal sources. Despite these inter-site differences, the relatively narrow range of overall mean concentrations (29.9–38.8 CFU/m³) suggests that the indoor environmental conditions in the reception areas of the six halls are broadly similar, likely reflecting the common building design standards and HVAC system specifications applied across the university's student accommodation facilities.

4.8 Antifungal Susceptibility Profiles

The antifungal susceptibility profiles of the four isolated fungal species are presented in Table 4. Zones of inhibition, measured in millimetres (mm), represent the diameter of the clear area around an antifungal disc where fungal growth has been inhibited. Larger zones indicate greater antifungal effectiveness. The results were interpreted according to CLSI breakpoint criteria as susceptible (S), intermediate (I), or resistant (R). A designation of susceptible (S) indicates that the fungus is inhibited at or below the breakpoint concentration of the drug; standard dosing is likely to be effective for treatment. A designation of intermediate (I) indicates that the fungus may respond to higher doses or more concentrated local applications of the drug, but standard dosing may not be sufficient. A designation of resistant (R) indicates that the fungus grows even at or above the breakpoint concentration; the drug is not effective for treatment.

Table 4: Antifungal susceptibility profiles of isolated fungal species (Kirby-Bauer disc diffusion method).

Organism	Ketoconazole	Fluconazole	Itraconazole	Griseofulvin
<i>A. niger</i>	S (18 mm)	R (0 mm)	S (20 mm)	I (12 mm)
<i>A. flavus</i>	S (16 mm)	R (0 mm)	S (17 mm)	I (11 mm)
<i>A. fumigatus</i>	S (15 mm)	R (0 mm)	S (19 mm)	S (14 mm)
<i>P. chrysogenum</i>	S (19 mm)	R (0 mm)	S (21 mm)	I (13 mm)

All four isolates were susceptible to both ketoconazole and itraconazole, with zone diameters ranging from 15 to 21 mm. The largest zone of inhibition for ketoconazole was observed with *P. chrysogenum* (19 mm), while the smallest was observed with *A. fumigatus* (15 mm). For itraconazole, the largest zone was again produced by *P. chrysogenum* (21 mm), while the smallest was produced by *A. flavus* (17 mm). Griseofulvin exhibited variable activity, with *A. fumigatus* being the only isolate classified as susceptible (14 mm), while the remaining three species showed intermediate susceptibility with zone diameters of 11–13 mm.

Notably, all four isolates exhibited complete resistance to fluconazole, with no measurable zone of inhibition (0 mm) observed for any of the tested species. This 100% resistance to fluconazole across all isolates is significant for public health because fluconazole is one of the most commonly prescribed azole antifungal agents for treating fungal infections. Complete resistance across all tested isolates indicates that this first-line therapeutic option would be ineffective for treating infections caused by these environmental isolates, necessitating the use of alternative, often more expensive or more toxic, antifungal agents. The clinical implications of this finding are particularly concerning given the widespread use of fluconazole in both community and hospital settings for the empiric treatment of suspected fungal infections.

5 Discussion

5.1 Discussion of Findings

5.1.1 Fungal Species Identified

Four fungal species were identified in the indoor air of the reception areas of the student halls of residence at the University of Cape Coast: *Aspergillus niger*, *Aspergillus flavus*, *Aspergillus fumigatus*, and *Penicillium chrysogenum*. The predominance of *Aspergillus* species among the indoor air isolates is consistent with the findings of numerous studies conducted in tropical and subtropical regions, where members of this genus are consistently reported as the most prevalent airborne fungi in both indoor and outdoor environments^{2, 3}. The identification of *A. niger* as the most prevalent species aligns with its well-documented cosmopolitan distribution and its ability to grow on a remarkably wide range of substrates under diverse environmental conditions.

The presence of *A. flavus* and *A. fumigatus* in the indoor air of the student halls is of particular clinical significance. *Aspergillus flavus* is a well-known producer of aflatoxins, which are among the most potent carcinogenic substances known, and its presence in indoor environments raises concerns about the potential for mycotoxin exposure through inhalation². *Aspergillus fumigatus* is the primary aetiological agent of invasive aspergillosis, a life-threatening infection with high mortality rates among immunocompromised patients². The isolation of this opportunistic pathogen from the indoor air of student accommodation facilities underscores the importance of maintaining indoor air quality standards, particularly in settings where immunocompromised individuals may be present.

The identification of *Penicillium chrysogenum* as a component of the indoor mycobiota is consistent with its well-established role as one of the most common indoor fungal contaminants worldwide². While *P. chrysogenum* is generally considered less pathogenic than the *Aspergillus* species identified in this study, it is a significant allergen and has been associated with respiratory symptoms in sensitised individuals. The small conidial size of *Penicillium* species (2.5–4 μm) enables efficient aerosolisation and deep penetration into the respiratory tract, potentially reaching the alveolar spaces and triggering both allergic and irritant responses.

5.1.2 Species Composition and Prevalence

The species composition observed in this study, dominated by *Aspergillus* and *Penicillium* species, is characteristic of indoor fungal communities in tropical climates and is consistent with the findings of similar studies conducted across West Africa and other tropical regions. The isolation of all three *Aspergillus* species from every sampling site suggests that these fungi are common and widespread components of the indoor mycobiota in the student halls of residence. The slightly lower prevalence of *Penicillium chrysogenum* (isolated from five of six sites) compared to the *Aspergillus* species may reflect differences in the ecological preferences of this genus, particularly its potentially greater sensitivity to the drying effects of the HVAC-conditioned indoor environment.

The taxonomic composition of the fungal aerosol in the sampled reception areas is noteworthy for the absence of certain genera that are commonly reported in indoor air studies, such as *Cladosporium* and *Alternaria*. These genera are typically associated with outdoor vegetation and are among the most abundant fungal spores in outdoor air, particularly during the growing season. Their absence from the indoor air samples in this study may be attributed to the effective filtration of outdoor air by the HVAC systems, which are designed to remove particulate matter from the incoming air stream before it enters the indoor environment. This observation supports the hypothesis that the HVAC systems in the student halls are playing a significant role in shaping the composition and concentration of the indoor fungal aerosol.

5.1.3 Estimated Fungal Concentrations and WHO Compliance

The estimated fungal concentrations in this study ranged from 21.0 to 49.3 CFU/m³, with all values falling below the WHO guideline limit of 50 CFU/m³ for indoor environments. This finding is significant as it indicates that the indoor air quality in the reception areas of all six student halls of residence at the University of Cape Coast is in compliance with international standards at the time of sampling. The highest recorded value of 49.3 CFU/m³ (Limpopo Hall, morning) approaches but does not exceed the WHO threshold, suggesting that while fungal concentrations are within acceptable limits, there is limited margin of safety, particularly during the morning hours when concentrations are at their peak.

The overall mean concentrations across the six sites (29.9–38.8 CFU/m³) are comparable to those reported in studies of indoor air quality in similarly managed institutional buildings with HVAC systems in tropical regions. Studies by Awosika et al.⁷ in Nigeria and by Medeiros et al.⁷ in Brazil reported indoor fungal concentrations in air-conditioned buildings that were generally below 50 CFU/m³, consistent with the findings of the present study. In contrast, studies conducted in naturally ventilated buildings in the same climatic zones have frequently reported concentrations exceeding the WHO guideline, highlighting the protective effect of HVAC systems in controlling indoor fungal burden.

The compliance of all sampled buildings with the WHO guideline is an encouraging finding that suggests the current building management practices, including the operation and maintenance of HVAC systems, are generally effective in controlling indoor fungal concentrations. However, it is important to note that the measurements represent a single sampling episode during the dry season, and fungal concentrations may vary significantly with seasonal changes in humidity and temperature. Continuous or periodic monitoring would be necessary to confirm that compliance is maintained throughout the year, particularly during the wet season when outdoor fungal concentrations are typically at their highest.

5.1.4 Temporal and Spatial Variation

The consistent observation of higher morning fungal concentrations compared to afternoon values across all six sampling sites is a key finding of this study. This diurnal pattern can be explained by the combined effects of HVAC filtration and the progressive

removal of airborne spores over the course of the day. The entrance doors of the reception areas are wide open at all times, and the windows are permanently closed because the air conditioning units are constantly operating. The HVAC systems' mechanical filters progressively trap airborne fungal spores throughout the day, and the continuous air recirculation through filter media reduces spore concentrations in the indoor environment.

In the morning, after a period of reduced or no HVAC operation during the night (or reduced effectiveness of filtration during periods of lower airflow), the indoor fungal spore concentration reflects the accumulation of spores that have entered the building through the open entrance doors and have not yet been removed by the filtration system. As the HVAC system operates continuously throughout the day, each cycle of air recirculation passes the indoor air through the filtration media, which physically trap a fraction of the airborne spores. This progressive filtration effect results in a gradual decline in the indoor fungal concentration, with the lowest values observed during the afternoon sampling period. The dilution effect contributed by the continuous exchange of air through the wide-open main entrance doors may also contribute to the afternoon reduction, as the influx of relatively spore-poor outdoor air during the afternoon hours can dilute the indoor fungal concentration.

The spatial variation observed across the six sampling sites, while not dramatic, does suggest that building-specific factors influence indoor fungal concentrations. Limpopo Hall consistently recorded the highest concentrations, while Valco Hall recorded the lowest. These differences may reflect variations in the volume and layout of the reception areas, the specific HVAC system configurations and filter specifications, the orientation and proximity of the buildings to outdoor fungal sources, and the occupancy levels during the sampling period. The relatively narrow range of overall mean concentrations (29.9–38.8 CFU/m³) across the six sites suggests that the fundamental conditions governing indoor fungal burden are similar across all sampled buildings, likely reflecting the common building standards and HVAC system designs employed in the university's student accommodation infrastructure.

5.1.5 Antifungal Susceptibility and Public Health Implications

The antifungal susceptibility testing revealed several findings of clinical and public health significance. The uniform susceptibility of all four isolates to ketoconazole and itraconazole is reassuring, as these antifungal agents represent important therapeutic options for the treatment of fungal infections caused by *Aspergillus* and *Penicillium* species. Itraconazole, in particular, is considered the drug of choice for the treatment of allergic bronchopulmonary aspergillosis and is widely used as first-line therapy for various forms of aspergillosis⁷. The susceptibility of all isolates to itraconazole indicates that this agent would likely be effective in treating infections caused by the environmental isolates identified in this study.

The complete resistance of all isolates to fluconazole (0 mm zone of inhibition) is the most concerning finding of the antifungal susceptibility analysis. Fluconazole is one of the most commonly prescribed azole antifungal agents for treating fungal infections, owing to its favourable pharmacokinetic profile, excellent oral bioavailability, low toxicity, and long half-life that permits once-daily dosing. Complete resistance across all tested isolates indicates that this first-line therapeutic option would be ineffective for treating

infections caused by these environmental isolates, necessitating the use of alternative, often more expensive or more toxic, antifungal agents. The clinical implications are significant: patients presenting with fungal infections potentially acquired from exposure to these environmental isolates would not respond to fluconazole therapy, potentially delaying effective treatment and increasing the risk of adverse outcomes.

The variable susceptibility to griseofulvin, with only *A. fumigatus* classified as susceptible while the other three species showed intermediate susceptibility, reflects the known limitations of griseofulvin as an antifungal agent. Griseofulvin is primarily effective against dermatophyte fungi and has limited activity against the opportunistic moulds identified in this study. The intermediate susceptibility observed for *A. niger*, *A. flavus*, and *P. chrysogenum* suggests that higher doses or more prolonged treatment courses might be required if griseofulvin were to be used, although this agent would not typically be considered a first-line treatment for infections caused by these organisms.

The observed fluconazole resistance may be related to the intrinsic resistance of *Aspergillus* species to this agent, as *Aspergillus* species are known to possess naturally low susceptibility to fluconazole due to structural differences in their target enzyme, lanosterol 14 α -demethylase, and the presence of efflux pumps that actively transport the drug out of the fungal cell⁷. However, the complete resistance of *P. chrysogenum* to fluconazole is more noteworthy, as *Penicillium* species generally exhibit greater susceptibility to azoles than *Aspergillus* species. This finding may reflect the acquisition of resistance mechanisms through environmental exposure to azole fungicides used in agriculture, a phenomenon that has been increasingly documented in environmental *Aspergillus* isolates and that may extend to other fungal genera.

5.2 Limitations of the Study

Several limitations of this study should be acknowledged. First, the settle plate method used for air sampling is a passive technique that does not measure a known volume of air, and the derived concentrations are estimates based on Omeliansky's formula, which makes assumptions about spore settling velocity that may not hold under all conditions. Active air sampling methods, such as the Andersen sampler or the SAS Super 180, would provide more accurate and reproducible concentration measurements but were not available for this study. Second, fungal identification was based solely on morphological criteria, which may not distinguish between closely related species or cryptic species within the *Aspergillus* and *Penicillium* genera. Molecular identification methods, such as PCR and DNA sequencing of the internal transcribed spacer (ITS) region, would provide more definitive species-level identification. Third, the study was conducted during a single season (dry season), and the results may not be representative of fungal concentrations during other seasons, particularly the wet season when higher humidity levels may support greater fungal growth and sporulation. Fourth, the study assessed antifungal susceptibility against only four agents, and the results may not reflect susceptibility patterns to other clinically important antifungal drugs such as amphotericin B, voriconazole, or the echinocandins. Fifth, the study did not assess the viability or infectivity of the airborne fungal spores, and the mere presence of fungal spores in the air does not necessarily indicate a health risk to building occupants. Finally, the cross-sectional design of the study limits the ability to draw conclusions about temporal trends or causal relationships between environmental factors and indoor fungal concentrations.

6 Conclusion and Recommendations

6.1 Conclusion

This study investigated the fungal microflora of indoor air in the reception areas of six student halls of residence at the University of Cape Coast, Ghana, during the dry season. Four fungal species were identified: *Aspergillus niger*, *Aspergillus flavus*, *Aspergillus fumigatus*, and *Penicillium chrysogenum*. The estimated fungal concentrations, calculated using Omeliansky's formula, ranged from 21.0 to 49.3 CFU/m³, with all values falling below the WHO guideline limit of 50 CFU/m³ for indoor environments. This finding indicates that the indoor air quality in the reception areas of all six student halls is in compliance with international standards at the time of sampling.

Morning fungal concentrations were consistently higher than afternoon values across all sampling sites, a pattern attributed to the continuous filtration of indoor air by Heating, Ventilation, and Air Conditioning (HVAC) systems operating throughout the day. The HVAC systems' mechanical filters progressively trap airborne fungal spores, and the continuous air recirculation through filter media reduces spore concentrations in the indoor environment over the course of the day. The highest recorded concentration (49.3 CFU/m³ at Limpopo Hall, morning) remains within the WHO guideline, though it approaches the threshold, indicating a limited margin of safety during peak exposure periods.

Antifungal susceptibility testing revealed that all isolates were susceptible to ketoconazole and itraconazole, variably susceptible to griseofulvin, and uniformly resistant to fluconazole. The complete resistance to fluconazole across all tested isolates is a significant public health concern, as this agent is one of the most commonly prescribed antifungal drugs and its ineffectiveness against environmental isolates limits therapeutic options for treating infections acquired from the indoor environment. These findings underscore the need for ongoing surveillance of indoor air quality and antifungal resistance patterns in institutional settings.

6.2 Recommendations

Based on the findings of this study, the following recommendations are made:

- (i) The university should implement a regular indoor air quality monitoring programme for student halls of residence and other institutional buildings, with sampling conducted during both the dry and wet seasons to capture seasonal variation in fungal concentrations.
- (ii) The maintenance schedules for HVAC systems in the student halls should be reviewed and strengthened to ensure that filters are replaced at appropriate intervals and that the systems are operating at optimal efficiency for the removal of airborne particulates, including fungal spores.
- (iii) Given the proximity of the highest recorded concentration (49.3 CFU/m³) to the WHO guideline of 50 CFU/m³, Limpopo Hall should be prioritised for enhanced air

quality monitoring and HVAC maintenance to ensure continued compliance with the standard.

- (iv) The university health service should be made aware of the antifungal susceptibility profiles of the environmental fungal isolates, particularly the complete resistance to fluconazole, to inform the empiric treatment of fungal infections among the student population.
- (v) Future studies should employ active air sampling methods and molecular identification techniques to provide more accurate and comprehensive data on indoor fungal concentrations and species diversity in the university's accommodation facilities.
- (vi) A prospective study assessing the clinical correlation between indoor fungal exposure and respiratory symptoms among hall residents would provide valuable data on the actual health impact of the fungal concentrations observed in this study.
- (vii) The use of higher-efficiency filters (such as HEPA filters) in the HVAC systems of the student halls should be considered as a means of further reducing indoor fungal concentrations and providing a greater margin of safety below the WHO guideline.

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Appendices

Appendix A: Omeliansky's Formula Derivation

Omeliansky's formula is used to convert settle plate colony counts into estimated airborne concentrations. The derivation is based on the assumption that spores settle from a column of air directly above the exposed plate. The volume of air from which spores are captured is determined by the surface area of the plate and the height of the air column, which is related to the exposure time and the assumed settling velocity of the spores.

The formula is:

$$\text{CFU/m}^3 = \frac{n \times 10,000}{\pi \times r^2 \times t} \quad (2)$$

where:

- n = number of colonies on the settle plate
- r = radius of the Petri dish (cm) = 4.5 cm for a standard 90 mm plate
- t = exposure time (minutes) = 15 minutes in this study
- 10,000 = conversion factor to express results in CFU/m³

Substituting the values used in this study:

$$\text{CFU/m}^3 = \frac{n \times 10,000}{3.14159 \times (4.5)^2 \times 15} = \frac{n \times 10,000}{954.26} = n \times 10.48 \quad (3)$$

Therefore, the estimated concentration in CFU/m³ equals the average colony count multiplied by a conversion factor of approximately 10.48.

Appendix B: CLSI Breakpoints for Antifungal Susceptibility

The following table presents the Clinical and Laboratory Standards Institute (CLSI) zone diameter breakpoints for the interpretation of antifungal disc diffusion test results, as specified in CLSI document M44-A2. These breakpoints are used to categorise isolates as susceptible (S), intermediate (I), or resistant (R) based on the measured zone of inhibition diameter.

Table 5: CLSI zone diameter breakpoints for antifungal disc diffusion (mm).

Antifungal Agent	Susceptible (S)	Intermediate (I)	Resistant (R)
Ketoconazole (10 µg)	≥15 mm	11–14 mm	≤10 mm
Fluconazole (25 µg)	≥19 mm	15–18 mm	≤14 mm
Itraconazole (10 µg)	≥15 mm	11–14 mm	≤10 mm
Griseofulvin (25 µg)	≥14 mm	10–13 mm	≤9 mm

Note: Zone diameter breakpoints for moulds are adapted from CLSI M44-A2 guidelines. For *Aspergillus* species, fluconazole is generally not considered an appropriate therapeutic agent, and intrinsic resistance is expected. The breakpoints listed for fluconazole are primarily applicable to yeast isolates but are included here for completeness and to contextualise the resistance pattern observed in this study.